Low-Code Experiment

Integrating a Zoho Creator Application with Another Zoho Service using Connections and APIs

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Abstract

This experiment studies how a Zoho Creator application can be integrated with other Zoho services. Zoho Creator uses **connections** (OAuth-based authorisation to another service) together with the Deluge **invokeUrl** task to call that service's REST API. For the HOTEL BOOKING application, an integration with Zoho Cliq was planned so that every new booking is announced in a Cliq channel, and the integration options available in the Creator version used in the lab were explored.

1. Aim

To understand and configure the integration of a Zoho Creator application with another Zoho service using connections and APIs.

2. Concepts

Connection. A saved, authorised link between Creator and another service. Zoho services use Zoho OAuth, where the user grants named scopes (for example Cliq webhooks and channels) and Creator stores and refreshes the token.

API call. The Deluge **invokeUrl** task sends an HTTP request (GET, POST, PUT, DELETE) to a service's API URL. When a connection name is supplied, the authorisation header is added automatically.

Prebuilt integrations. Creator also provides ready-made connectors in the workflow action picker, which let the user choose an action without writing code.

3. Integration Design

Use case. When a guest submits the Room Booking Form, the front desk should be notified immediately in a Zoho Cliq channel named hotelbookings, in addition to the confirmation email sent to the guest in Experiment 6.

Trigger. Form workflow on Room Booking Form: Created > Successful form submission.

Procedure.

1. In Zoho Cliq, create an organisation and a channel named hotelbookings.
2. In Creator, create a connection named cliqconn using Zoho OAuth with the Cliq Webhooks (CREATE) and Channels (READ) scopes, and authorise it.
3. In the existing workflow, add a Deluge script action that builds the message from the submitted record and posts it with invokeUrl.
4. Submit a test booking and check the message in the channel.

Deluge script

```
checkIn = input.Check_in_date.toString("dd-MMM-yyyy");
msg = "New booking: " + input.Guest_Name.first_name + " | " + input.Room_Type
```

```
      + " | " + input.Number_Of_Guests + " guests | Check-in " + checkIn;
resp = invokeUrl
[
    url : "https://cliq.zoho.in/api/v2/channelsbyname/hotelbookings/message"
    type : POST
    parameters : {"text" : msg}.toString()
    connection : "cliqconn"
];
info resp;
```

4. Implementation Observations

4.1 Zoho Cliq setup. Signing in to Zoho Cliq with the institute account first asks the user to join an existing organisation on the same email domain or to create another company. A separate lab company was created for this experiment, rather than joining an unrelated organisation.

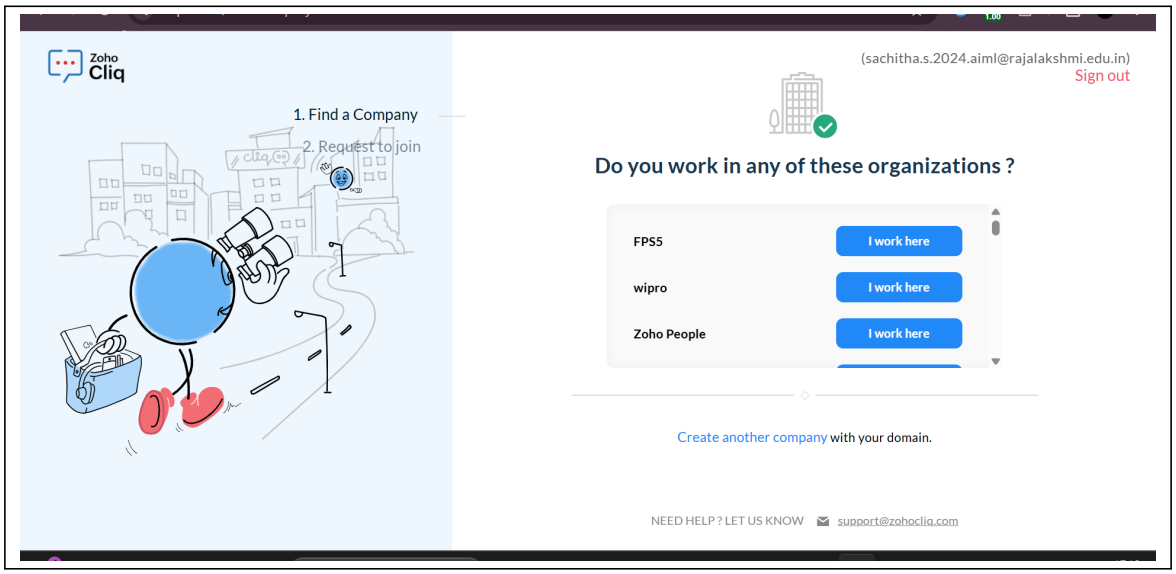


Figure 1: Zoho Cliq organisation selection on first sign-in

Screenshot to be inserted: hotelbookings channel created in Cliq

4.2 Where connections are managed. The application Settings page in this Creator version lists Permissions, Personalization and Developer Tools (Schema Builder, Application IDE, Widgets, Variables). There is no separate Connections page here, so a connection is created from inside the Deluge editor, through the Connection option of the Invoke URL task.

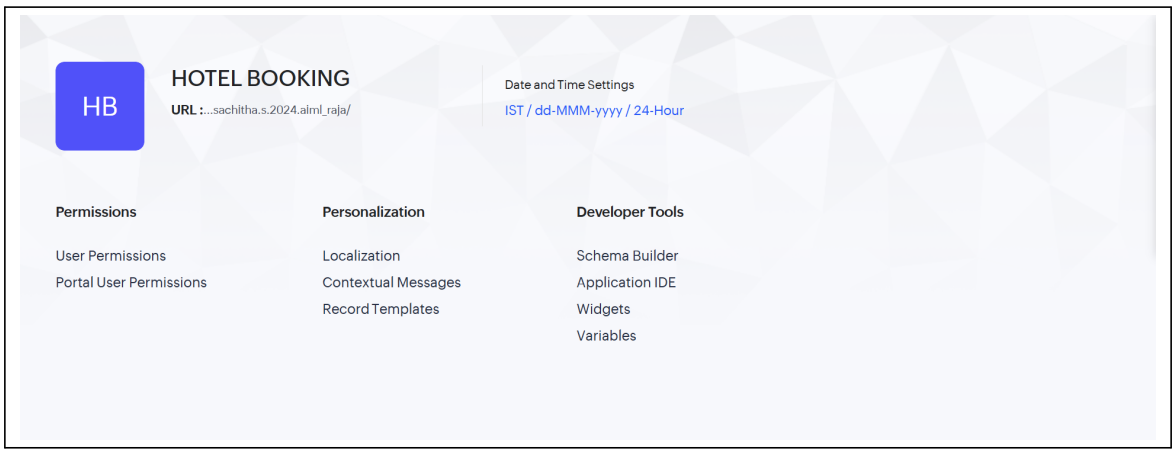


Figure 2: Application Settings page of HOTEL BOOKING

4.3 Prebuilt connectors. The Integrations tab of the workflow action picker offers QuickBooks, SalesForce, Zoho CRM, Zoho Recruit and SAP OData as ready-made connectors. Zoho Cliq is not in this list, which is why the Cliq integration uses a Deluge script with invokeUrl instead of a prebuilt action.

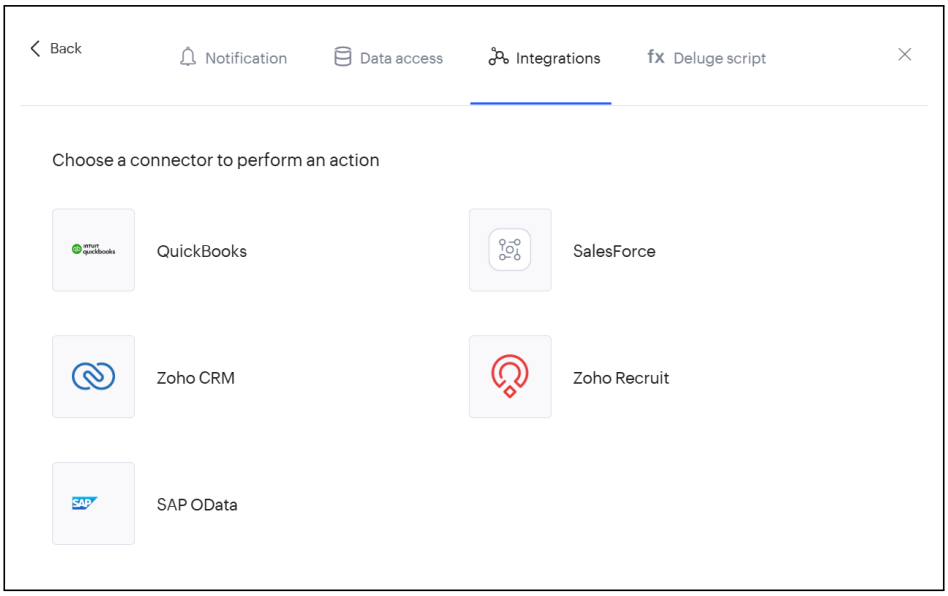


Figure 3: Integrations tab in the workflow action picker

Screenshot to be inserted: connection cliqconn authorised and connected

Screenshot to be inserted: booking message received in the Cliq channel

5. Result

The connection and API approach for integrating HOTEL BOOKING with Zoho Cliq was designed, and the integration options available in the lab's Creator version were identified. The end-to-end test output is to be added in the marked screenshot boxes after the connection is authorised.

6. Learning Outcome

Learnt how Zoho Creator connects to other Zoho services using OAuth connections and the invokeUrl task, how prebuilt connectors differ from API calls written in Deluge, and how the available integration options depend on the Creator version and plan.